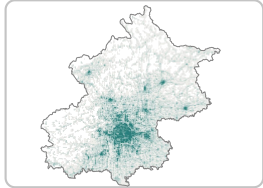


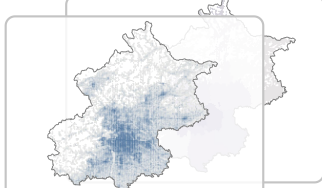
Data

1 km grid-year panel · Beijing
12,236 populated grids ·
2018-2024



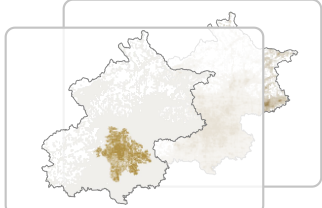
veterinary & comparison POIs

+



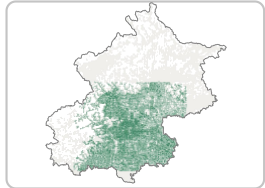
population & housing market

+



buildings & nighttime lights

+



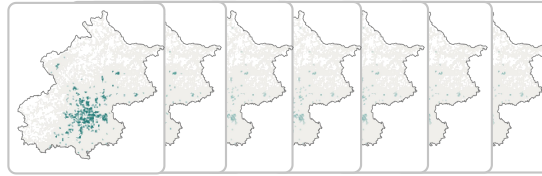
OSM roads & mobility

1

Step 1: Exposure landscape and service benchmarks

● **RQ (i): how did local pet-healthcare exposure evolve?**

• Population-weighted exposure — Eq. (1)

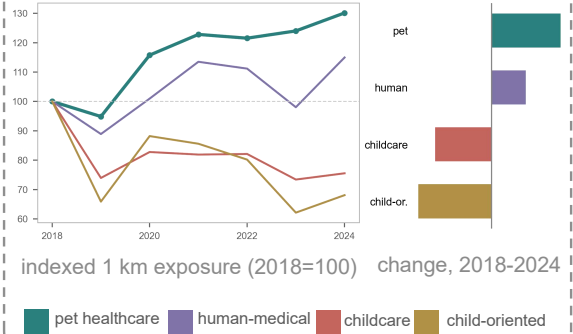


annual exposure surfaces, 2018 -> 2024

Measure: veterinary sites within reach of each 1 km grid,
weighted by resident population

Radii: 500 m · 1 km (main) · 2 km

• Cross-service trajectories, 2018-2024



2

Step 2: Demand signals, morphology and serviceability

● **RQ (ii)-(iii): are demand signals observable, stable — and separable?**

• Demand signal D — Eqs. (2)-(3)



stacked inputs -> composite demand
signal D

Inputs: three layers · seven indicators
(homes · pet traces · family care)

Validation: entry · platform · specificity
ablation · scale

• Urban morphology M — Eq. (5)

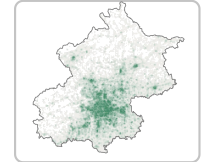


four form blocks -> composite morphology M

Blocks: activity · built form ·
functional mix · intensity

Checks: exact-GIS overlays
(correlation > 0.97)

• Serviceability model — Eq. (6)



serviceability score S

Layers: medical · pet services ·
family substrate · parking

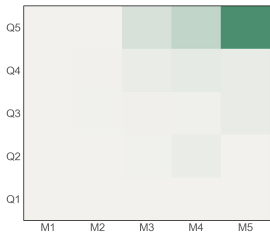
Tests: D, M and S correlated,
but separable

3

Step 3: Conversion gradient, supply-exposure mismatch and mechanisms

● **RQ (iv): where do high demand signals become local supply?**

• Mismatch state — Eq. (4)



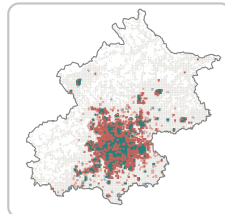
matched supply by demand x morphology

high-demand stratum: matched vs latent

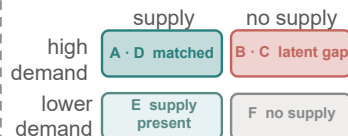
Stratum: top demand quintile, split
by local 1 km supply

Pattern: matched share rises with
morphology

• Mismatch typology & clustering

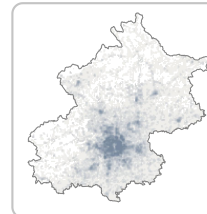


mismatch typology, six classes



Tests: Moran's I & LISA on five spatial
layers

• Mechanism diagnostics

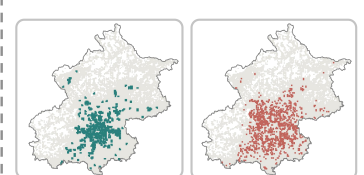


parking affordance (mobility layer)

Mechanisms: service ecology · metro x
parking · street spillover

Models: XGBoost + SHAP ·
LPM (district FE)

• Outcome



Claim: demand converts where
morphology, services and
mobility align

Object: exposed but unserved
care landscapes

Evidence: Figs. 2-7 · Tables 1 -6